

KARAGANDY, KazakhstanWMO: **353940**Lat: **49.671N**Lon: **73.334E**Elev: **1765**StdP: **13.78**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-21.8	-34.4	0.8	-27.7	-27.6	1.3	-21.1	33.6	21.4	30.6	20.9	5.6	70	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	89.8	62.3	86.3	61.4	82.8	60.3	65.3	80.9	63.9	79.3	62.7	77.9	8.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.8	85.0	68.7	59.0	79.7	67.5	57.3	74.7	67.2	31.2	80.7	30.1	79.5	29.2	77.4	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	28.6	24.3	20.9	DB	-32.2	95.7	6.6	3.8	-37.0	98.4	-40.8	100.6	-44.5	102.7	-49.3	105.5
(5)				WB	-32.2	68.9	6.5	2.2	-36.9	70.5	-40.7	71.8	-44.4	73.0	-49.1	74.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	5.3	7.6	21.8	42.8	54.7	64.4	66.4	63.9	52.7	39.9	22.6	10.3
	DBStd	24.63	13.02	12.73	12.52	10.45	8.38	6.96	6.22	7.58	9.09	9.54	13.54	13.02
	HDD50	6217	1385	1188	876	256	51	2	0	3	73	331	822	1230
	HDD65	10177	1850	1608	1340	666	330	92	56	112	379	777	1272	1695
	CDD50	1791	0	0	1	41	198	434	509	434	155	19	0	0
	CDD65	276	0	0	0	1	11	74	100	79	11	0	0	0
	CDH74	4290	0	0	0	30	267	1151	1328	1233	276	5	0	0
	CDH80	1444	0	0	0	4	58	375	479	447	81	0	0	0
Wind	WSAvg	8.6	8.8	9.3	10.2	9.1	9.1	8.1	7.5	7.4	7.5	8.3	8.9	9.5
Precipitation	PrecAvg	14.1	0.9	0.8	1.0	1.1	1.5	1.5	2.1	1.0	0.7	1.1	1.2	1.1
	PrecMax	19.9	1.7	2.0	2.4	3.0	3.9	4.0	5.0	2.6	1.5	3.3	2.5	3.1
	PrecMin	9.1	0.2	0.2	0.2	0.1	0.4	0.1	0.3	0.0	0.0	0.3	0.2	0.4
	PrecStd	3.0	0.4	0.4	0.6	0.7	0.8	1.2	1.3	0.8	0.4	0.7	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	35.7	57.7	78.5	86.4	91.8	94.7	93.5	88.2	73.7	54.8	36.2
		MCWB	32.8	33.4	46.0	54.3	59.6	63.0	64.5	62.9	59.3	50.9	44.4	33.9
	2%	DB	31.2	32.2	47.8	71.5	80.4	87.8	89.5	89.3	81.2	67.3	49.0	33.3
		MCWB	29.6	30.5	41.4	52.2	57.5	62.0	63.3	61.8	55.8	48.9	42.1	31.5
	5%	DB	27.2	28.8	41.3	66.2	76.6	84.3	85.4	85.2	76.3	61.9	43.2	30.4
		MCWB	25.9	27.1	37.4	49.8	55.7	60.8	62.3	60.8	54.7	46.8	39.1	28.4
	10%	DB	23.6	26.2	37.5	61.0	73.0	80.8	81.2	80.8	71.4	56.2	38.7	27.2
		MCWB	22.2	24.7	34.5	47.4	54.2	60.0	61.4	59.5	52.8	44.4	35.2	25.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	33.7	47.0	56.2	62.2	66.5	67.5	66.4	61.0	53.0	47.8	34.6
		MCDB	34.7	35.2	55.8	72.0	79.5	83.1	83.7	82.7	82.7	67.5	50.8	35.6
	2%	WB	29.7	30.5	41.4	53.5	59.4	64.7	65.6	64.0	58.1	50.2	42.9	31.5
		MCDB	31.0	31.9	46.9	68.9	74.3	81.0	80.6	81.0	76.6	63.9	48.3	32.6
	5%	WB	26.0	27.3	37.6	51.0	57.6	63.2	64.4	62.4	55.8	48.0	39.0	28.5
		MCDB	27.1	28.6	41.0	63.2	72.1	78.8	79.0	80.0	72.9	59.7	43.7	30.0
	10%	WB	22.3	24.4	34.6	48.4	55.8	61.7	63.2	61.1	53.6	45.4	35.2	25.7
		MCDB	23.5	25.9	37.0	59.2	69.4	76.6	77.2	78.1	68.9	54.5	38.3	27.1
Mean Daily Temperature Range	5% DB	MDBR	17.8	19.2	17.7	22.6	26.3	25.3	23.8	26.4	26.6	21.2	16.3	17.0
		MCDBR	13.5	16.8	19.3	31.6	32.8	31.8	31.1	32.2	34.4	32.7	20.5	14.9
		MCWBR	12.9	15.7	13.8	15.7	15.0	12.8	11.9	12.5	15.4	17.8	14.6	13.8
	5% WB	MCDBR	13.4	16.0	18.2	27.9	27.8	27.1	26.0	28.1	31.3	28.4	19.2	14.2
		MCWBR	13.1	15.3	13.4	15.0	13.9	12.4	11.2	12.3	15.1	16.4	14.5	13.7
Clear-Sky Solar Irradiance	taub	0.291	0.325	0.376	0.408	0.409	0.411	0.436	0.400	0.354	0.339	0.325	0.296	
	taud	2.067	1.998	1.986	2.131	2.190	2.250	2.186	2.271	2.352	2.303	2.149	2.065	
	Ebn at Noon	228	249	257	264	270	269	260	264	266	246	213	205	
	Edh at Noon	27	38	46	45	44	42	44	38	32	28	25	24	
All-Sky Solar Radiation	RadAvg	505	880	1291	1570	1976	2088	1902	1748	1336	752	394	360	
	RadStd	71	42	99	119	168	143	134	125	96	102	36	48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-187	-67
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-187	-67

Nomenclature: See separate page